

Cambridge (CIE) IGCSE Maths
Paper 4: Calculator (Set C)
Extended

Saturday 26 April 2025

Morning (Time: 2 hours 0 minutes)

Total marks

/ 102

Instructions

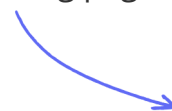
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- You should use a scientific calculator where appropriate.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 In triangle ABC , $BC = 7.6\text{cm}$ and $AC = 6.2\text{ cm}$.

Using a ruler and compasses only, construct triangle ABC .

Leave in your construction arcs.

The side AB has been drawn for you.



(2 marks)

- 2 Simplify.

$$2x^2 \times 5x^5$$

(2 marks)

- 3 (a)** Louise leaves home at 09 55 and cycles the 5.6 km to the supermarket at a constant speed.
She takes 15 minutes to complete the journey.

Write down the time she arrives at the supermarket.

[1]

(1 mark)

- (b)** Calculate Louise's average speed from her home to the supermarket

i) in kilometres per hour,

.....km/h [1]

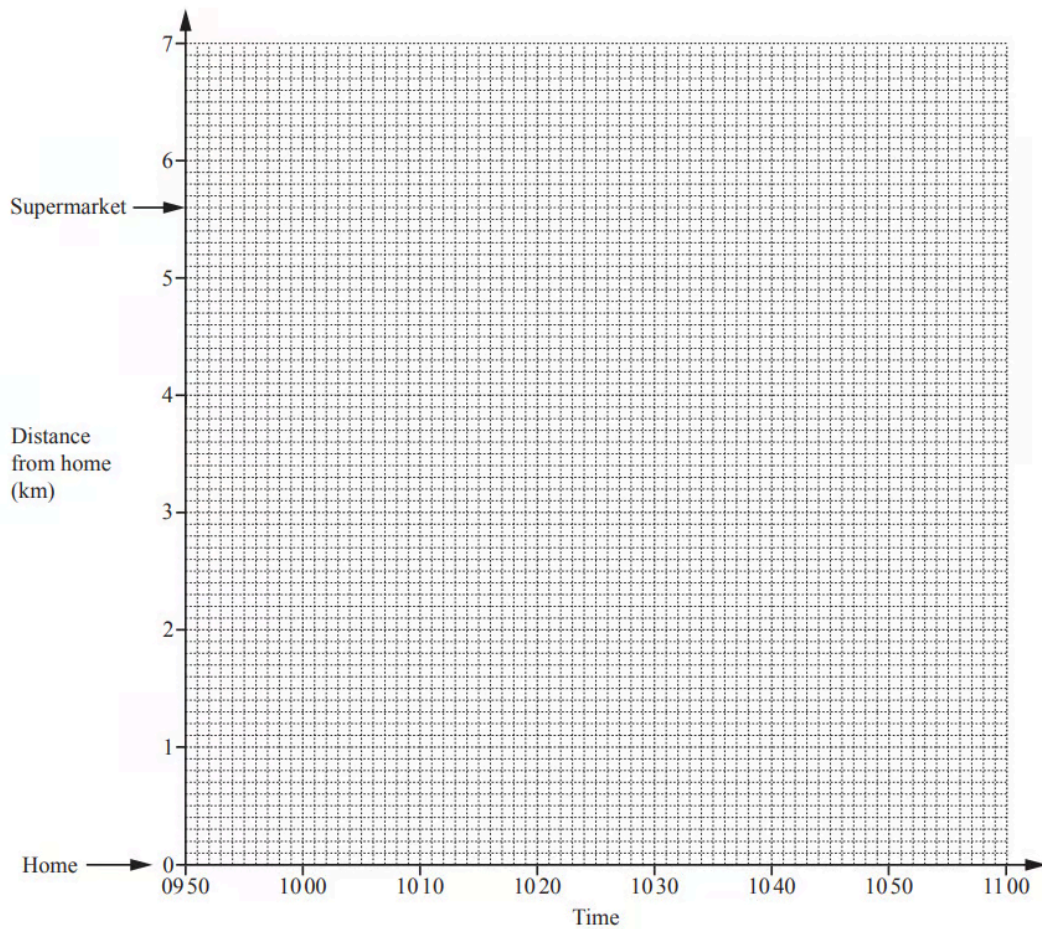
ii) in metres per second, giving your answer correct to 1 decimal place.

..... m/s [2]

(3 marks)

- (c)** Louise stays at the supermarket for 23 minutes.

On the grid, draw the travel graph of her journey from home and her stay at the supermarket.



(2 marks)

- (d) Louise's mother leaves home at 10 07 to meet Louise at the supermarket. She cycles at a constant speed of 28 km/h.

i) Work out how long she takes for the 5.6 km journey. Give your answer in minutes.

.....min [2]

ii) On the grid, show her mother's journey.

[1]

(3 marks)

(e) They cycle home together at a constant speed and arrive at 1054.

i) On the grid, show their journey home.

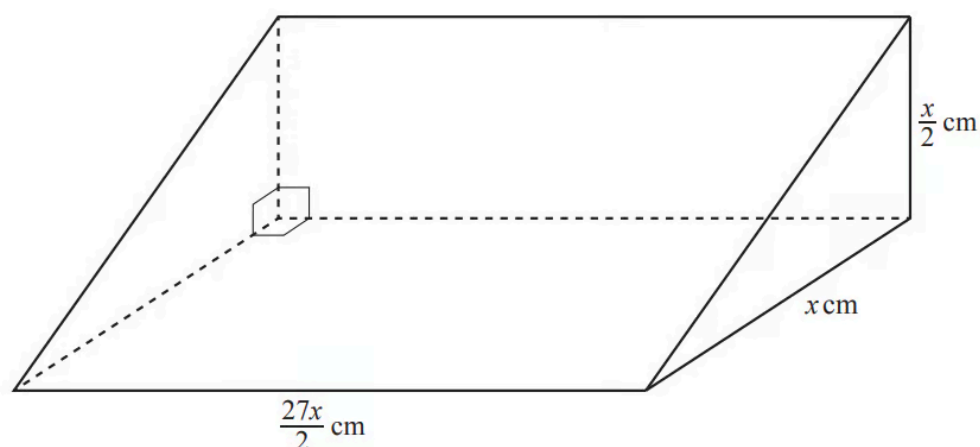
[1]

ii) Calculate, in km/h, their constant speed on the journey home.

.....km/h [2]

(3 marks)

4



The diagram shows a prism with a right-angled triangle as its cross-section. The volume of the prism is 1000 cm^3 .

Calculate the value of x .

$x = \dots\dots\dots$

(4 marks)

- 5** A bag contains red marbles, green marbles and blue marbles only.
The ratio of the number of marbles of each colour is

red : green : blue = 12 : 5 : 2.

There are 112 more red marbles than green marbles.

Work out the number of blue marbles.

(2 marks)

- 6** \$8400 is invested for 2 years at a rate of 3.5% per year compound interest.
Work out the total amount of interest earned by the end of the 2 years.

\$.....

(3 marks)

7 i) Find the highest common factor (HCF) of 36 and 90.

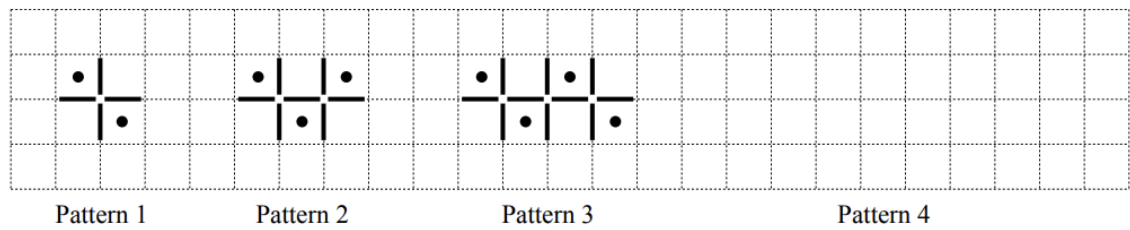
[2]

ii) Find the lowest common multiple (LCM) of 36 and 90.

[2]

(4 marks)

8 (a) A sequence of patterns is made using lines and dots. The first three patterns in the sequence are shown below.



Draw Pattern 4 on the grid.

(1 mark)

(b) Complete the table.

Pattern	1	2	3	4		10
Number of dots	2	3				
Number of lines	4	7				

(4 marks)

(c) Find an expression, in terms of n , for

i) the number of dots in Pattern n ,

[1]

ii) the number of lines in Pattern n .

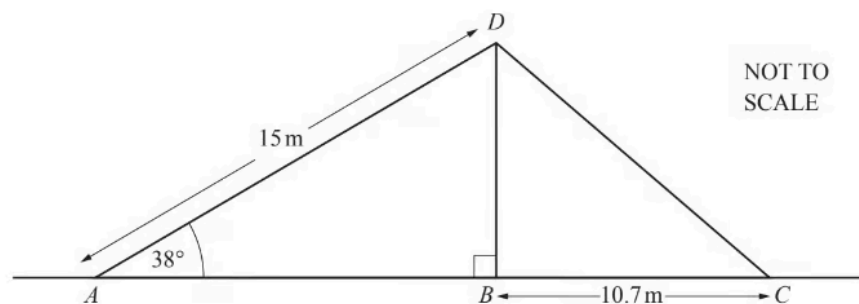
[2]

(3 marks)

(d) A pattern has 76 lines. Work out how many **dots** are in this pattern.

(2 marks)

9 (a)



A vertical flagpole, BD , stands on horizontal ground and is held by two ropes, AD and CD .

$AD = 15\text{m}$, $BC = 10.7\text{m}$ and angle $DAB = 38^\circ$.

Using trigonometry, calculate BD .

$BD = \dots\dots\dots\text{m}$

(2 marks)

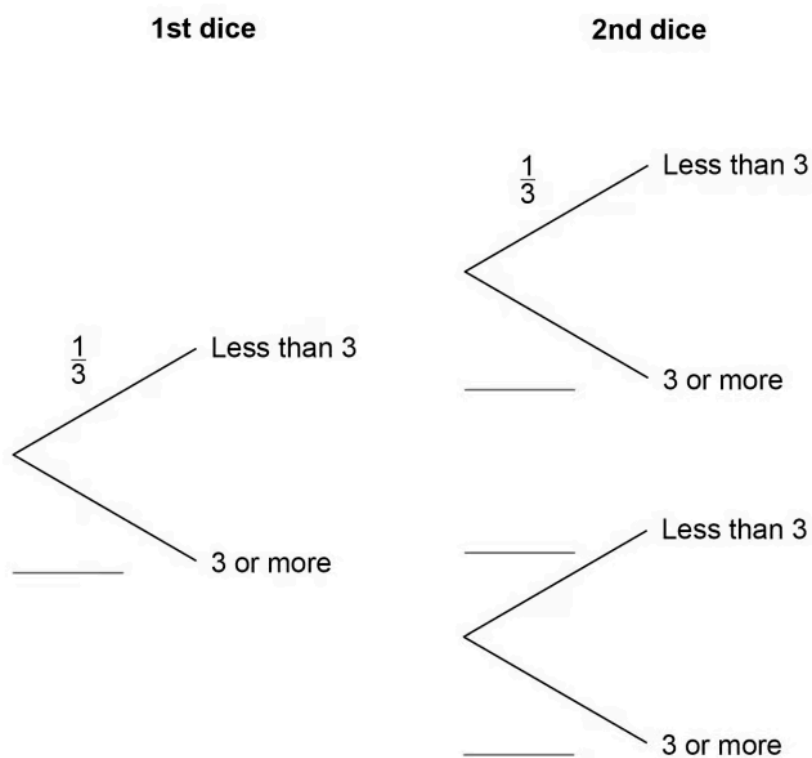
(b) Calculate CD .

$CD = \dots\dots\dots\text{m}$

(2 marks)

10 (a) Two ordinary fair dice are rolled.

Complete the tree diagram.



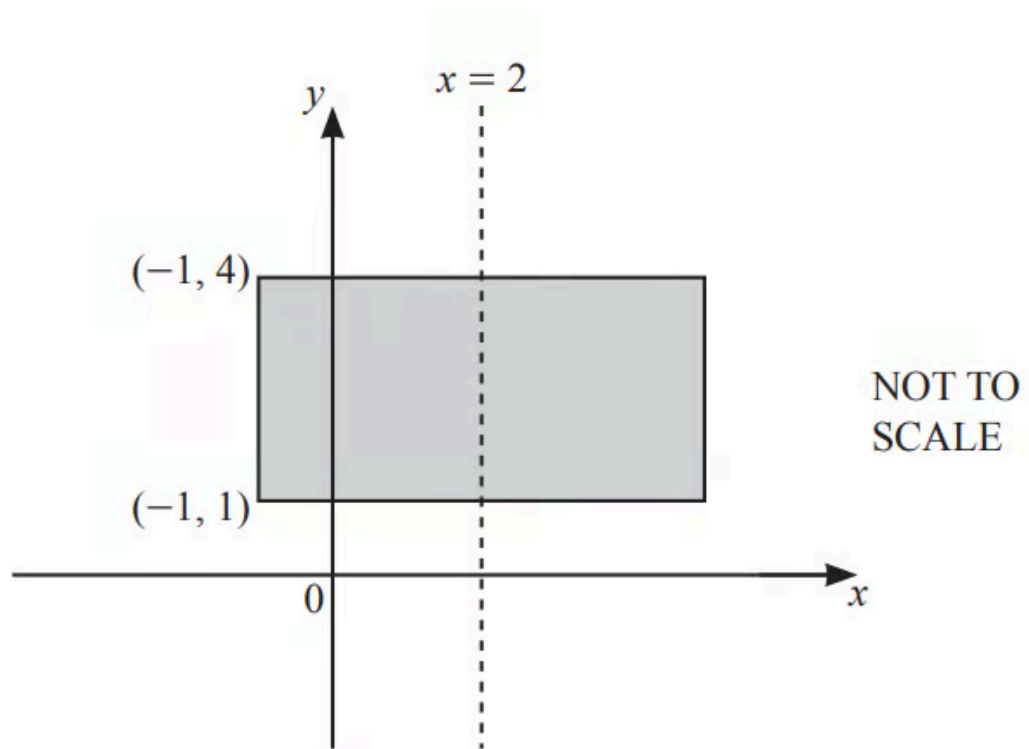
(1 mark)

(b) Work out the probability that **both** dice land on a number less than 3.

(1 mark)

(c) Work out the probability that **exactly one** of the dice lands on a number less than 3.

(2 marks)



The diagram shows a rectangle with a line of symmetry at $x = 2$. Two vertices of the rectangle are at $(-1, 1)$ and $(-1, 4)$.

The shaded region is defined by the inequalities $a \leq x \leq b$ and $c \leq y \leq d$.

Find the values of a , b , c and d .

$a =$

$b =$

$c =$

$d =$

(2 marks)

12 Factorise.

$$xy - 2x + 3y - 6$$

(2 marks)

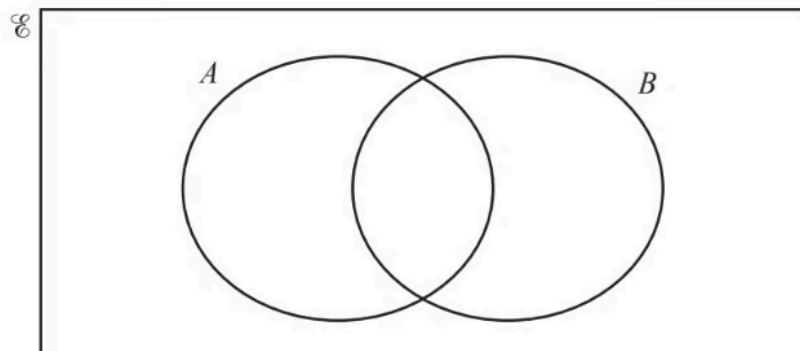
13 (a) $\mathcal{E} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$

$A = \{x : x \text{ is even}\}$

$4 \in A \cap B$

$n(A \cap B) = 1$

$(A \cup B)' = \{1, 7, 9\}$



Complete the Venn diagram using this information.

(4 marks)

(b) Use your Venn diagram to complete the statement.

$B = \{ \dots \}$

(1 mark)

14 Bob asked each of 40 friends how many minutes they took to get to work.
The table shows some information about his results.

Time taken (m minutes)	Frequency
$0 < m \leq 10$	3
$10 < m \leq 20$	8
$20 < m \leq 30$	11
$30 < m \leq 40$	9
$40 < m \leq 50$	9

Work out an estimate for the mean time taken.

(4 marks)

15

$$s = ut + \frac{1}{2}at^2$$

Rearrange the formula to write a in terms of u , t and s .

$a =$

(3 marks)

16 (a) A is the point $(8,5)$ and B is the point $(-4,1)$.

Calculate the length of AB .

(3 marks)

(b) Find the co-ordinates of the midpoint of AB .

(..... ,)

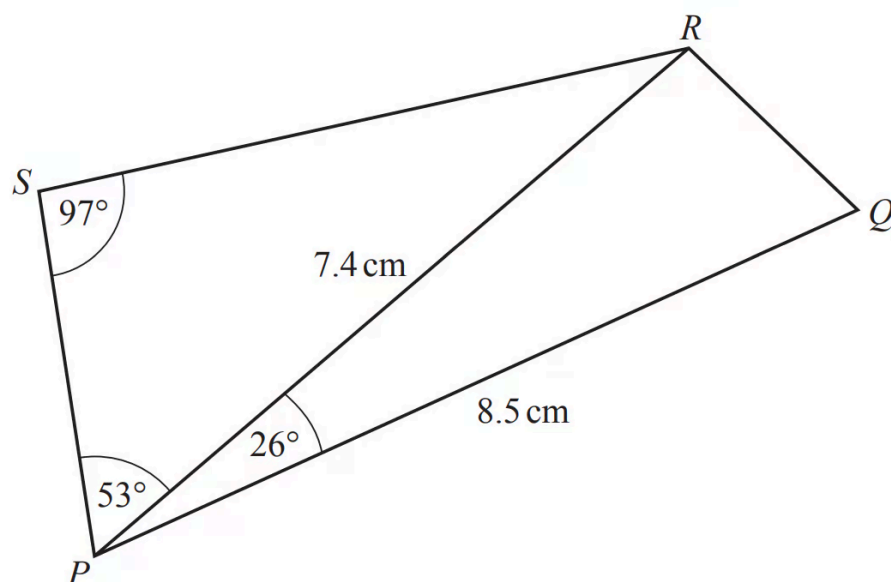
(2 marks)

17 The coordinates of P are $(-3, 8)$ and the coordinates of Q are $(9, -2)$.

Find the equation of the perpendicular bisector of PQ .

(4 marks)

18 (a)



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Calculate SR .

$SR = \dots\dots\dots\text{ cm}$

(3 marks)

(b) Calculate RQ .

$RQ = \dots\dots\dots\text{ cm}$

(4 marks)

19 A sphere has radius $x\text{ cm}$. The volume of the sphere is 1000 cm^3 .

Calculate the value of x .

[The volume, V , of a sphere with radius r is $V = \frac{4}{3} \pi r^3$.]

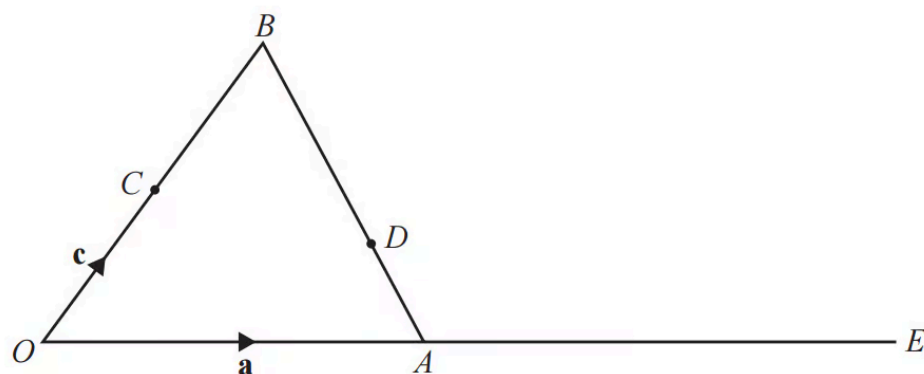
$x = \dots\dots\dots$

(3 marks)

20 Differentiate $6 + 4x - x^2$.

(2 marks)

21 (a)



OAB is a triangle and C is the mid-point of OB .

D is on AB such that $AD:DB = 3:5$.

OAE is a straight line such that $OA:AE = 2:3$.

$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.

Find the following vectors, in terms of $\mathbf{a}$ and $\mathbf{c}$, in their simplest form

i) $\vec{AB}$,

$\vec{AB} = \dots\dots\dots$ [1]

ii) $\vec{AD}$,

$\vec{AD} = \dots\dots\dots$ [1]

iii) $\vec{CE}$,

$\vec{CE} = \dots\dots\dots$ [1]

iv) $\vec{CD}$.

$\vec{CD} = \dots\dots\dots$ [2]

(5 marks)

(b) $\vec{CE} = k\vec{CD}$

Find the value of k .

$$k = \dots\dots\dots$$

(1 mark)

- 22** A road is 4530 m long, correct to the nearest 10 metres. Kirsty drove along the road in 205 seconds, correct to the nearest 5 seconds.

The average speed limit for the road is 80 km/h.

Could Kirsty's average speed have been greater than 80 km/h? You must show your working.

(5 marks)

23 Simplify.

$$\frac{2x^2 - x - 1}{2x^2 + x}$$

(4 marks)

24 Solve the equation $\frac{2x+5}{3-x} = \frac{14}{15}$.

$x = \dots\dots\dots$

(3 marks)